

Similarity and normalisation

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Embeddings and neighbourhoods

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Augmentation and leakage

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compare the rankings for Similarity and normalisation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Similarity and normalisation. Proposed explanation: “Cosine is a probability”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Compare a new case independently

Use query $[5, 0]$ and candidates $P [10, 0]$, $R [4, 3]$ and $S [0, 5]$. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates. Explain the changed ranking.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace the unit vectors

For $Q [3, 4]$, $A [6, 8]$ and $C [-3, -4]$, give each length and unit vector. Explain one distinction that normalisation removes and one that it preserves.

My reasoning and evidence

6. W2 Compare the rankings

For query Q , write the raw squared distances and cosines to A , B and C from the lesson. Name the winner under each rule. Calculate squared distances after normalising both sides and check the relationship to cosine.

My reasoning and evidence

7. W3 Handle a vector without direction

A candidate set contains Z [0, 0] and B [3, 0]. With nonzero query Q, what happens under this lesson's zero-vector policy? What if Q is zero, or Z is the only candidate? Explain why replacing an undefined cosine with zero is misleading.

My reasoning and evidence

8. W4 Compare a new case independently

Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates. Explain the changed ranking.

My reasoning and evidence

9. W5 Audit a proposed improvement

For event-count vectors $[3, 4]$ and $[6, 8]$, what useful information can normalisation remove? A team chooses cosine because it improves results on the final evaluation set, then calls that same set independent evidence. Explain how to repair the evaluation plan.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled similarity and normalisation evidence sheet to Image-pattern Network Investigation. Show how you can calculate vector length and normalise a nonzero vector. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

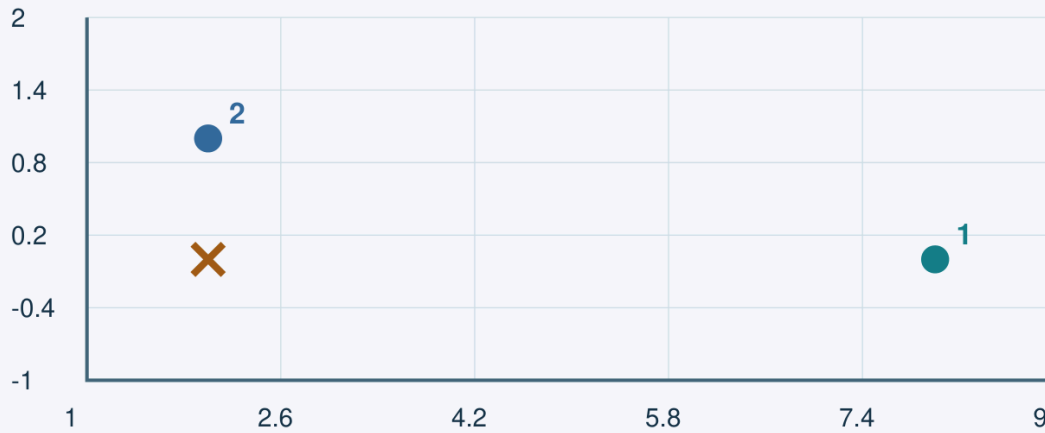
12. Pictorial case practice: Similarity and normalisation

Compare angular alignment with raw distance on the same vectors.

Query $[2, 0]$; $P=[8, 0]$; $Q=[2, 1]$. Nonzero vectors only; compare squared Euclidean distance and cosine.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Raw proximity differs from angular alignment



Horizontal: first vector coordinate; vertical: second vector coordinate

1: P at (8, 0)

2: Q at (2, 1)

Cross: query at (2, 0)

Distance rule: Compare squared Euclidean distance and cosine

Similarity and normalisation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Raw distances

P distance 36; Q distance 1, so Q is closer in raw coordinates.



2. Angular similarity

$\text{Cosine}P=1$; $\text{cosine}Q=2/\sqrt{5}\approx 0.8944$, so P aligns more closely.



3. Normalise

Unit query $[1, 0]$, P $[1, 0]$, Q $[2/\sqrt{5}, 1/\sqrt{5}]$; normalised distance P0.

CHANGE ONE THING

Replace P with $[0, 0]$.

Explain the changed result using the matching lesson.

Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare angular alignment with raw distance on the same vectors.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Unit query $[1, 0]$, $P [1, 0]$, $Q [2/\sqrt{5}, 1/\sqrt{5}]$; normalised distance PQ .

1. Angular similarity
2. Normalise
3. Raw distances

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Normalise Unit query $[1, 0]$, $P [1, 0]$, $Q [2/\sqrt{5}, 1/\sqrt{5}]$; normalised distance PQ . Raw Euclidean distance includes differences in both direction and magnitude. Cosine compares direction using the vector lengths, so a long vector can align perfectly while being far away in raw coordinates. Declare which question your similarity measure answers. A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

Practice 2: explain the deciding evidence

Changed-case practice: Replace P with $[0, 0]$. Which conclusion is supported by the supplied facts?

1. Cosine and unit normalisation for P are undefined.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Cosine and unit normalisation for P are undefined. The zero vector has norm 0, so dividing by its norm is invalid.

Repair a missing building block

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
